[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

ELECTRICAL DISCHARGE MACHINE, MANUAL AND COMPUTER NUMERICAL CONTROL (CNC)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers manual and computer numerical control (CNC) electrical discharge machines (EDMs) intended for use in manufacturing molds, stamping dies, and extrusion dies.

2. CLASSIFICATION. The EDMs shall be of the following types. The type and size to be furnished shall be as specified (see 7.2(b)).

Type I - Manually controlled machine
Type II - Computer numerical controlled machine
Type III - Z-axis computer numerical controlled machine

Size - See table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3410

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TABLE I. EDM sizes.

Type	Size	Minimum machine travels (inches)		
		X-axis	Y-axis	Z-axis
I	1	6	5	4
	2	11	7	5
	3	13	11	7
	4	19	11	7
	5	39	23	15
II	6	7	4	3
	7	11	7	5
	8	12	8	7
	9	15	9	7
	10	19	13	12
	11	22	15	9
	12	25	17	12
	13	31	19	13
	14	39	23	17
	15	51	23	15
	16	79	26	18
III	17	5	5	4
	18	11	7	7
	19	13	11	7
	20	19	14	9
	21	39	23	15

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The electrical discharge machine shall include, as a minimum, the following components:

3.2.1 Worktable. Unless otherwise specified (see 7.2(c)), the worktable length, width, weight-holding capacity and maximum distance from the worktable to platen shall be that considered as standard by the manufacturer. Unless otherwise specified (see 7.2(d)), the size, number, and configuration of worktable T-slots shall conform to the manufacturer's standard practices.

3.2.2 Automatic tool changer. If required for type II machines, an automatic tool changer shall be provided as specified (see 7.2(e)).

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3.2.3 Dielectric system. The EDM shall be equipped with a recirculating dielectric system capable of supplying dielectric to the cutting area and maintaining stable operating temperatures. The machine shall allow selection of manual or automatic control over the flow of dielectric. The system shall include cartridge-type filters. The dielectric shall be cooled by a thermostatically controlled chilling system that shall regulate the temperature of the dielectric at the workpiece to within 2 degrees of ambient temperature while the machine is cutting. The dielectric system shall filter particles to a 5 micron level. Unless otherwise specified, the internal dimensions of the dielectric working tank shall be that considered as standard by the manufacturer (see 7.2(f)).

3.2.4 Manual control requirements. Unless otherwise specified (see 7.2(g)), type I EDMSs shall include, as a minimum, the following features:

a. Spark gap control.
b. Pulse on/off time control.
c. Emergency stop button.
d. Power on/off switches.
e. Dielectric system pump on/off.

3.2.5 CNC requirements. Unless otherwise specified (see 7.2(h)), type II EDMs shall be capable of continuous path, 3-axis (X, Y, and Z), linear, circular, and helical interpolation and shall provide adaptive control by monitoring the spark gap parameters and pulse width in response to machining conditions. When specified (see 7.2(i)), type II EDM machines shall have C-axis simultaneous control. Type III EDMs shall be capable of continuous machining along the Z-axis. The CNC system shall provide adaptive control of machining operations by regulating current and spark gap in response to machining conditions.

3.2.5.1 Hardware features. Unless otherwise specified (see 7.2(j)), the CNC system shall include, as a minimum, the following hardware features:

a. Video display terminal.
b. 250-MB hard disk drive.
c. 3.5-inch floppy disk drive.
d. Electronic Industries Association (EIA) RS-232 interface.
e. Keypad for manual data input.

3.2.5.2 Control panel. Unless otherwise specified (see 7.2(k)), the CNC system shall include, as a minimum, the following control features:

a. Emergency stop button.
b. Power on/off switches.
c. Jog and jog direction.

3.2.5.3 Operating modes. Unless otherwise specified (see 7.2(l)), the CNC system shall include, as a minimum, the following operating modes:

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a. Manual data input.
b. Conversational/interactive mode - The conversational mode shall provide the operator with the ability to load an existing program, create a new program, and edit an existing program by following on-screen prompts. As a minimum, the conversational mode shall include prompts for axis and tool management data and part geometry data for creating a part program.
c. Machine diagnostics and monitoring.
d. Dry run (type II only).
e. Part program editing.

3.2.5.4 Axis and electrode management. Unless otherwise specified (see 7.2(m)), the CNC system shall include, as a minimum, the following axis and tool management features:

a. Electrode length compensation.
b. Electrode radius compensation.
c. Backlash compensation.
d. Offset machining.
e. Horizontal, 3-D, and side servomachining (type II only).
f. Orbital machining.

3.2.5.5 Software features. Unless otherwise specified (see 7.2(n)), the CNC system shall include, as a minimum, the following software programming and control features:

a. Linear, circular, and helical interpolation (type II only).
b. Inch/metric switchable programming.
c. Floating decimal point.
d. 3-D graphic display (type II only).
e. Automatic zero point return.
f. Standard G codes, M codes, and canned cycles.
g. Incremental and absolute coordinate positioning.
h. Scaling function for X and Y axes (type II only).

3.2.5.6 Optional software. Optional software shall be furnished as specified (see 7.2(o)).

3.2.6 Electrical requirements. Unless otherwise specified (see 7.2(p)), the electrical system shall conform to the National Fire Protection Association (ANSI/NFPA) 79. The machine shall have single 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(q)). Unless otherwise specified (see 7.2(r)), the CNC system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply. The average machining current capacity of the power supply shall not be less than specified (see 7.2(s)).

3.2.7 Electric motors. The EDM drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical

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Manufacturers Association (NEMA) MG-7. Unless otherwise specified (see 7.2(t)), the motors shall have open dripproof enclosures.

3.2.8 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(u)).

3.3 Performance. Unless otherwise specified (see 7.2(v)), the following performance requirements shall be meet for the machine type specified. Type I and III EDMs shall meet the International Standards Organization (ISO) 230-1 requirements specified in table II. Type III EDMs having a single column shall meet the ISO 11090-1 performance requirements specified in table III. Type III EDMs having two columns shall meet the ISO 11090-2 performance requirements specified in table IV. Testing shall be in accordance with sections 5.3 through 5.3.3.

TABLE II. Type I and III EDM performance requirements.

ISO 230-1 paragraph	Parameter	Tolerance, inch (millimeter)
5.322 5.324	Flatness of work-holding surface [per 36 in. (914 mm.) of travel]	0.001 (0.025)
5.232.1 5.232.11	Straightness of X-axis movement [per 12 in. (305 mm.) of travel]	0.0005 (0.013)
5.232.1 5.232.11	Straightness of Y-axis movement [per 12 in. (305 mm.) of travel]	0.0005 (0.013)
5.512.3	Squareness between X-axis movement and Y-axis movement [per 12 in. (305 mm.) of travel]	0.0005 (0.013)
5.522.2	Squareness between Z-axis movement and X-axis movements [per 12 in. (305 mm.) of travel]	0.0005 (0.013)
5.522.2	Squareness between Z-axis movement and Y-axis movements [per 12 in. (305 mm.) of travel]	0.0005 (0.013)
5.322.1	Rotary spindle square to table 0°, 90°, 180°, 270° [per 12 in. (305 mm.) from spindle]	0.0005 (0.013)
5.2112 5.212.1	Center T-slot parallel with longitudinal table travel [per 12 in. (305 mm.) of travel]	0.001 (0.025)
5.422.21 5.422.22	Parallelism between electrode platen and X-axis movement [per 12 in. (305 mm.) of travel]	0.001 (0.025)
5.422.21 5.422.22	Parallelism between electrode platen and Y-axis movement [per 12 in. (305 mm.) of travel]	0.001 (0.025)

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TABLE III. Type II, single-column performance requirements.

Test name ISO 11090-1	Parameter	Tolerance, inch (millimeter)
G1	X-axis straightness in XY and ZY planes	0.00059 (0.015)
G2	Y-axis straightness in XY and YZ planes	0.00059 (0.015)
G3	X- and Y-axis squareness	0.00059 (0.015)
G4	Squareness between Z-axis and X and Y motions	0.00079 (0.02)
G5	Squareness between W-axis and X and Y motions	0.00059 (0.015)
G6	Quill roll during Z-axis motion	0.00157 (0.04)
G7	Table flatness	0.00118 (0.03)
G8	Parallelism between table and X and Y axes	0.00059 (0.015)
G9	T-slot or table straightness in X-direction	0.00079 (0.02)
G10	T-slot or table parallelism in X-direction	0.00059 (0.015)
G11	Parallelism between platen and X and Y axes	0.00059 (0.015)
G12	Spindle bore run-out near spindle nose and at 3.937 in. (100 mm.)	0.00004/0.00039 (0.001/0.01)
G13	Parallelism between spindle and Z-axis in ZX and YZ planes	0.00039 (0.01)
G14	Backlash between quill and table in X and Y directions	0.00157 (0.04)
P1	X-, Y-, and Z-axis backlash (manually operated axes)	0.00118 (0.03)
P2	X-axis accuracy, repeatability, and reversal value	0.00043/0.00020/0.00031 (0.011/0.005/0.008)
P3	Y-axis accuracy, repeatability, and reversal value	0.00043/0.0002/0.00031 (0.011/0.005/0.008)
P4	Z-axis accuracy, repeatability, and reversal value	0.00035/0.00016/0.00024 (0.009/0.004/0.006)
P5	C-axis accuracy, repeatability, and reversal value	80/40/40 arcseconds
M1	90° and 120° pitch accuracy and diameter difference of holes	0.00079, 0.00118/0.00079 (0.02, 0.03/0.02)

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TABLE IV. Type II, two-column performance requirements.

Test name ISO 11090-2	Parameter	Tolerance, inch (millimeter)
G1	Table flatness	0.00118 (0.03)
G2	X-axis straightness in XY and ZX planes	0.00059 (0.015)
G3	Y-axis straightness in XY and YZ planes	0.00059 (0.015)
G4	Parallelism between table and X and Y axes	0.00079 (0.02)
G5	Reference slot or surface straightness in X-direction	0.00079 (0.02)
G6	Reference slot or surface parallelism of X-axis movement	0.00157 (0.04)
G7	X- and Y-axis squareness	0.00079 (0.02)
G8	Squareness between Z-axis and X and Y motions	0.00079 (0.02)
G9	Squareness between W-axis and X and Y motions	0.00079 (0.02)
G10	Quill roll during Z-axis motion	0.00079 (0.02)
G11	Parallelism between table and X and Y axes	0.00118 (0.03)
G12	Spindle bore run-out near spindle nose and at 3.937 in. (100 mm)	0.00020/0.00039 (0.005/0.01)
G13	Parallelism between spindle and Z-axis in ZX and YZ planes	0.00039 (0.01)
G14	Backlash between quill and table in X and Y directions	0.00079 (0.02)
G15	Backlash in axes feed mechanism (manual machine only)	0.00118 (0.03)
N1	X-axis accuracy, repeatability, and reversal value	0.00063/0.00051/0.00016 (0.016/0.013/0.004)
N2	Y-axis accuracy, repeatability, and reversal value	0.00063/0.00051/0.00016 (0.016/0.013/0.004)
N3	Z-axis accuracy, repeatability, and reversal value	0.00051/0.00039/0.00018 (0.013/0.010/0.003)
N4	C-axis accuracy, repeatability, and reversal value	120/80/20 arcseconds
P1	90° and 120° pitch accuracy and diameter difference of holes	0.00079, 0.00118/0.00079 (0.02, 0.03/0.02)

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3.4 Electrode weight capacity. Unless otherwise specified (see 7.2(w)), the electrode weight capacity of the EDM shall be that considered as standard by the manufacturer.

3.5 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(x)).

3.6 Safety and health requirements. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910.

3.7 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(y)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(z)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7.1 Dials. Unless otherwise specified (see 7.2(aa)), all dials indicating axis positions shall be graduated in increments of 0.001 inch.

3.8 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(bb)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amps, phase, frequency).
f. Short circuit/overcurrent rating.
g. Contract number or order number.
h. National stock number or plant equipment code.
i. Date of manufacture.

3.9 Lubrication plate or chart. When specified (see 7.2(cc)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The EDM shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(dd)), the EDM shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(ee)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(ff)). Unless otherwise specified (see 7.2(gg)), the machine shall be tested in accordance with the environmental conditions described in section 3 of ISO 230-2.

5.3.2 Operational test. The EDM and its CNC system shall be operated in accordance with the manufacturer's standard operating procedures for warm-up and run-off checks. The machine shall be operated at no load for not less than 1 hour. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. For type I and III EDMs, the performance tests shall be conducted in accordance with ISO 230-1. For type II EDMs, the performance tests shall be conducted in accordance with ISO/FDIS 11090-1 or ISO/FDIS 11090-2. The test results will be compared to the performance levels listed in the table II, III, or IV as applicable.

5.4 Acceptance. Unless otherwise specified (see 7.2(hh)), preliminary and final acceptance testing shall be conducted at the manufacturer's site. The results of the performance tests will be compared to the performance criteria listed in tables II through IV as applicable. Any failure of the machine to meet the criteria listed in tables II through IV shall be cause for rejection.

6. PACKAGING. Preservation, packing, and marking shall be as specified (see 7.2(ii)).

7. NOTES

7.1 Source of documents.

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7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following address:

Electronic Industries Association (EIA)

RS-232	Interface Between Data Terminal Equipment and Data Circuit Terminating Equipment Employing Serial Binary Data Interchange

Application for copies should be addressed to the Electronic Industries Association, 2500 Wilson Boulevard, Arlington, VA 22201-3834.

International Standards Organization (ISO)

ISO/FDIS 11090-1	Machine tools - Test conditions for die-sinking, electrodischarge machines (EDM) - Terminology and testing of the accuracy, Part 1: Single column machines
ISO/FDIS 11090-2	Machine tools - Test conditions for die-sinking, electrodischarge machines (EDM) - Terminology and testing of the accuracy, Part 2: Two column machines
ISO 230-1	Test Code for Machine Tools - Part 1: Geometric Accuracy of Machines Operating Under No-Load or Finishing Conditions
ISO/DIS 230-2	Test code for machine tools - Part 2: Determination of accuracy and repeatability of positioning of numerically controlled axes

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

National Electrical Manufacturer's Association

NEMA MG-7	Motion/Position Control Motors and Controls

Applications for copies should be sent to the National Electrical Manufacturer's Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

ANSI/NFPA-79	Electrical Standard for Industrial Machinery

Applications should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

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Occupational Safety and Health Administration (OSHA)

OSHA 1910 General Industry, Safety, and Health Standards (29 CFR 1910)

Applications for copies should be sent to the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type and size of EDM required (see 2).
c. Worktable dimensions, weight-holding capacity, and maximum distance to platen, if different (see 3.2.1).
d. Worktable T-slots, if different (see 3.2.1).
e. Automatic tool changer, as specified (see 3.2.2).
f. Working tank internal dimensions, if different (see 3.2.3).
g. Manual controls, if different (see 3.2.4).
h. CNC requirements, if different (see 3.2.5).
i. C-axis simultaneous control, if required (see 3.2.5).
j. Hardware features, if different (see 3.2.5.1).
k. Control panel, if different (see 3.2.5.2).
l. Operating modes, if different (see 3.2.5.3).
m. Axis and tool management, if different (see 3.2.5.4).
n. Software features, if different (see 3.2.5.5).
o. Optional software, as specified (see 3.2.5.6).
p. Electrical system industrial standard, if different (see 3.2.6).
q. Electrical system voltage, as specified (see 3.2.6).
r. Control system voltage, if different (see 3.2.6).
s. Average machining current capacity, as specified (see 3.2.6).
t. Motor enclosures, if different (see 3.2.7).
u. Optional accessories, as specified (see 3.2.8).
v. Performance, if different (see 3.3).
w. Electrode weight capacity, if different (see 3.4).
x. Machine dimensions and weight, if different (see 3.5).
y. System of units required (see 3.7).
z. Dual system of units, if required (see 3.7).
aa. Dial increment graduations, if different (see 3.7.1).
bb. Nameplate, if different (see 3.8).
cc. Lubrication plate or chart, if required (see 3.9).
dd. Testing, if different (see 5.3).
ee. Test responsibility, if different (se 5.3.1).
ff. Certification of calibration, as specified (see 5.3.1)
gg. Test environmental conditions, if different (see 5.3.1).
hh. Acceptance, if different (see 5.4).
ii. Packaging requirements, as specified (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

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Size: See table I

Type: I - Manually controlled machine
II - Computer numerical controlled machine
III - Z-axis computer numerical controlled machine

CID number

MILITARY INTERESTS:

Custodians
Army - AL
Air Force - 99
Navy - SH

Reviewers
Air Force - 84
Navy - MC, AS, OS
Army - CR

PREPARING ACTIVITY:
DLA - GS

(Project 3410-0054)

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